

Preliminary Results of Gaze Tracking and Hand Motor Synergy Coordination During Reach-to-Grasp VR Tasks for Improved Prosthetic Arm Intent Prediction



Eitan Gerber, Nili E. Krausz

Neuro-Robotics and Bionic Limbs (eNaBLE) Lab

Faculty of Mechanical Engineering, Technion



Introduction

- Upper limbs have over 30 DOFs (arm+hand) in total [1]
- Amputees are **unable** to coordinate multi-joint prostheses
- **Motor synergies** could provide a predictive input for full arm semi-autonomous control.
- We propose leveraging Gaze and EMG synergies to improve prediction *timing* and *accuracy* (see Krausz et al.[2])

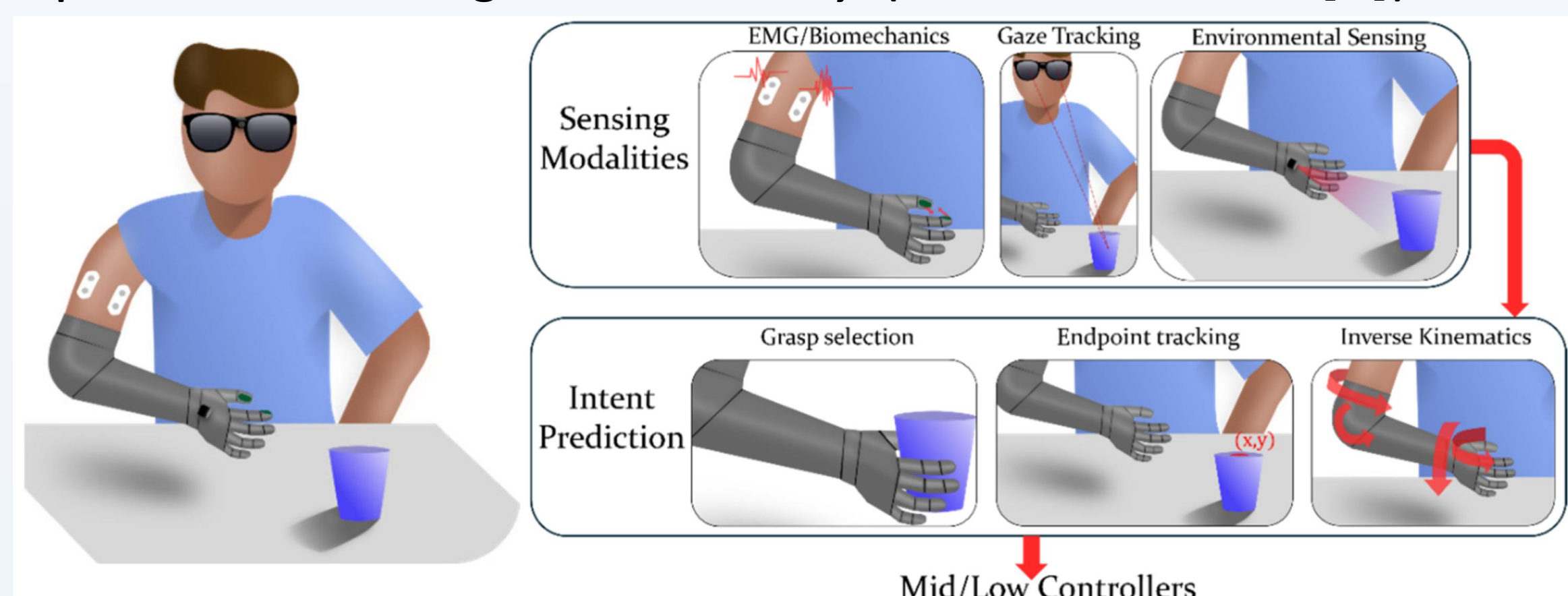


Fig 1. Multi-modal semi-autonomous prosthesis control concept from MOST BIONAC project

Methods

- Analyzed data from the 3D-ARM-Gaze dataset [3]
- Hand and gaze **Velocities** and **Accelerations** were computed
- Position, Velocity and Acceleration were filtered via 5Hz LPF
- Gaze-point motion was explored in a head reference frame
- Hand motion was explored in a head-gaze reference frame

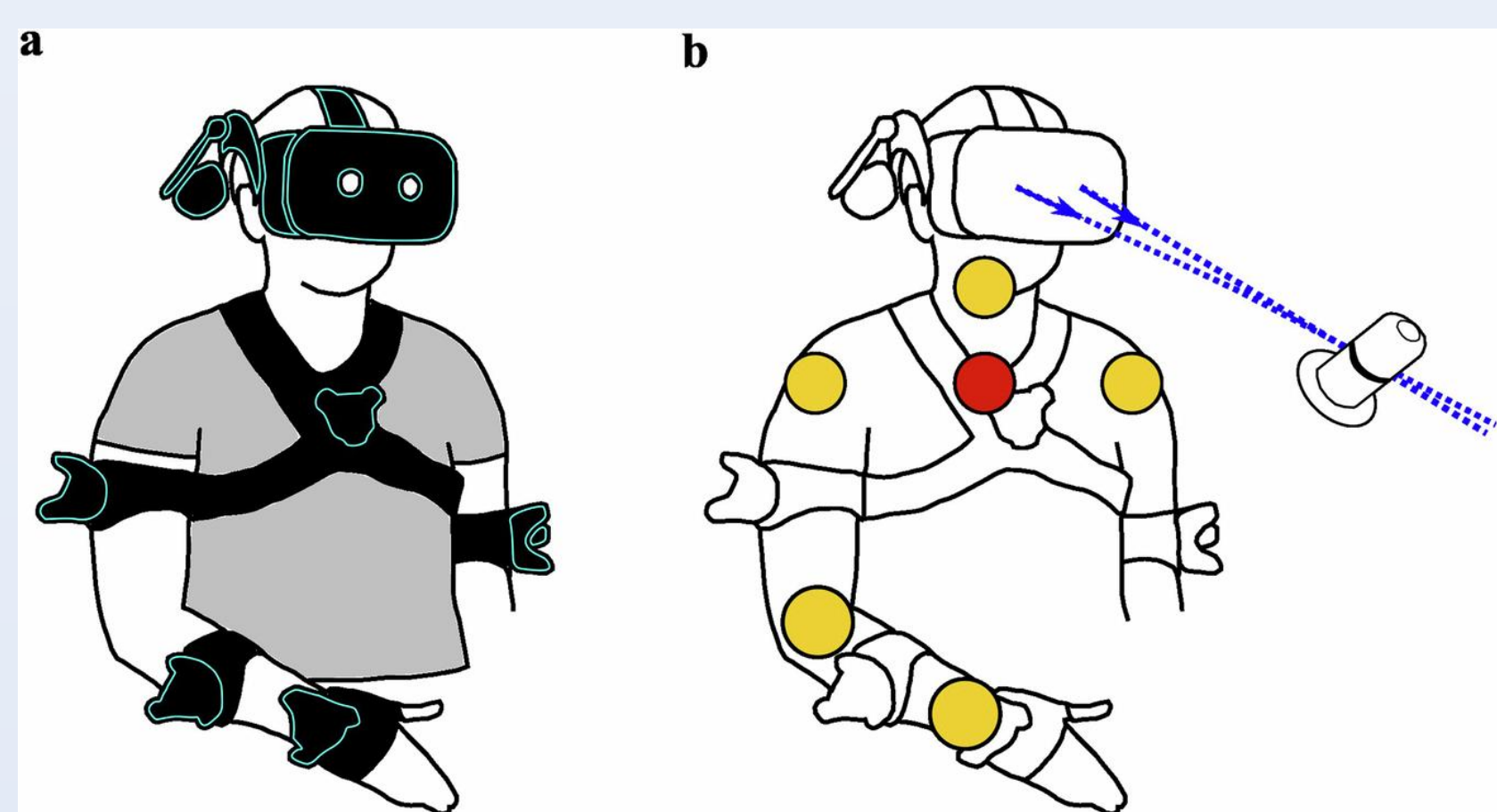


Fig 4. Illustration of the 3D-ARM dataset's collection method.

- a) The VR headset and MoCap trackers.
- b) Estimated joint centers and gaze vectors.

Adapted from [3]

Future Work

- Develop **integrated platform** for recording multi-modal gaze, surface electromyography (sEMG) and kinematics.
- Conduct “in-house” study including naturalistic activities in real-world settings (versus VR).
- We want to better understand muscle, postural, and gaze synergies during upper limb movements.
- Early motor synergies will be leveraged for **predicting user intent**.
- These will be implemented in a prosthetic arm controller.

Preliminary Results

- Gaze point peak acceleration **anticipates** hand peak acceleration at the start of reaching tasks, by $\sim 147[ms]$.

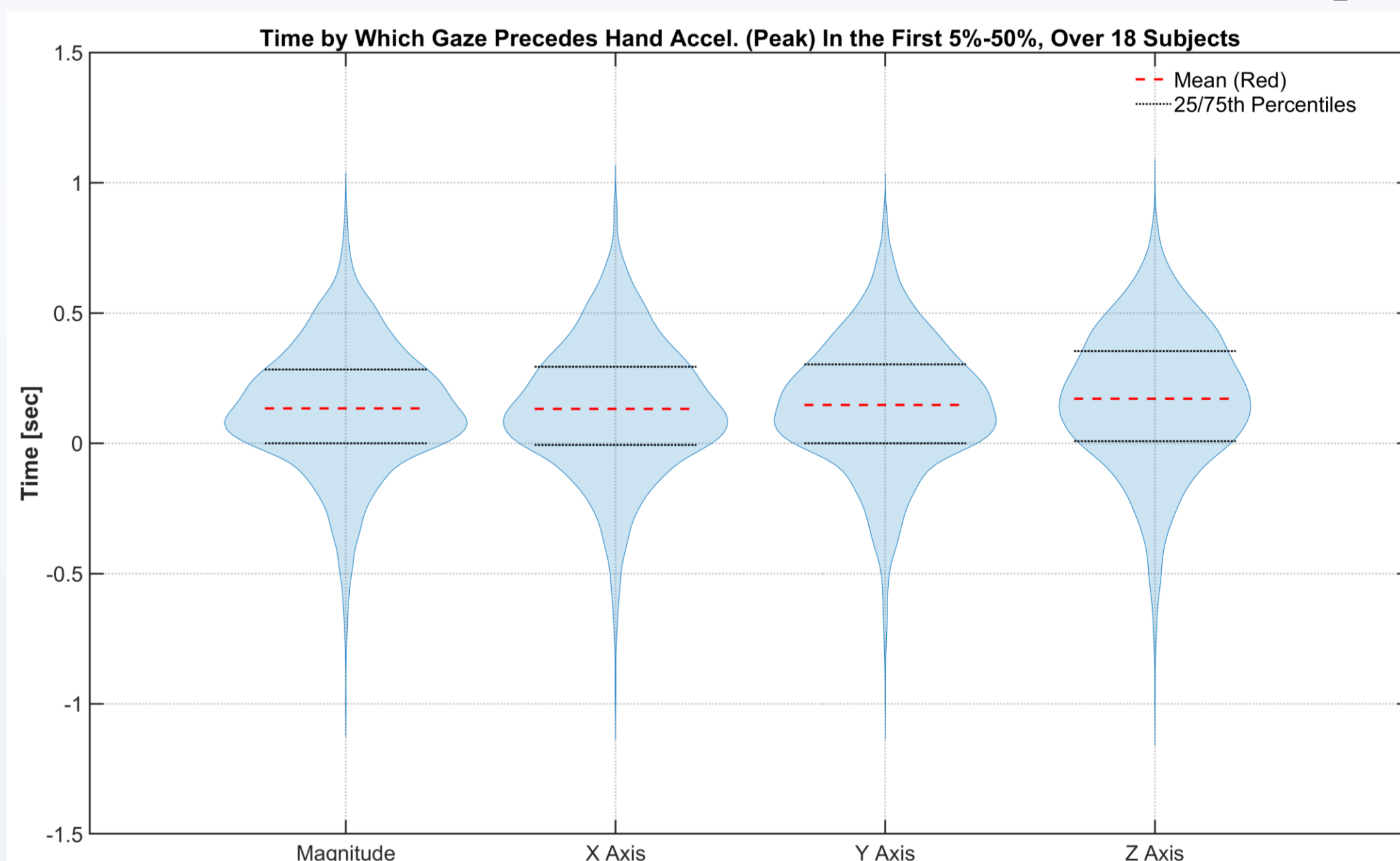


Fig 2. Distribution of time between gaze acceleration and hand acceleration peaks

- Gaze position begins to converge $\sim 30\%$ into a reaching task, yet remains offset.
- Hand position converges slower, surpassing gaze accuracy $\sim 50\%$ into a reaching task.

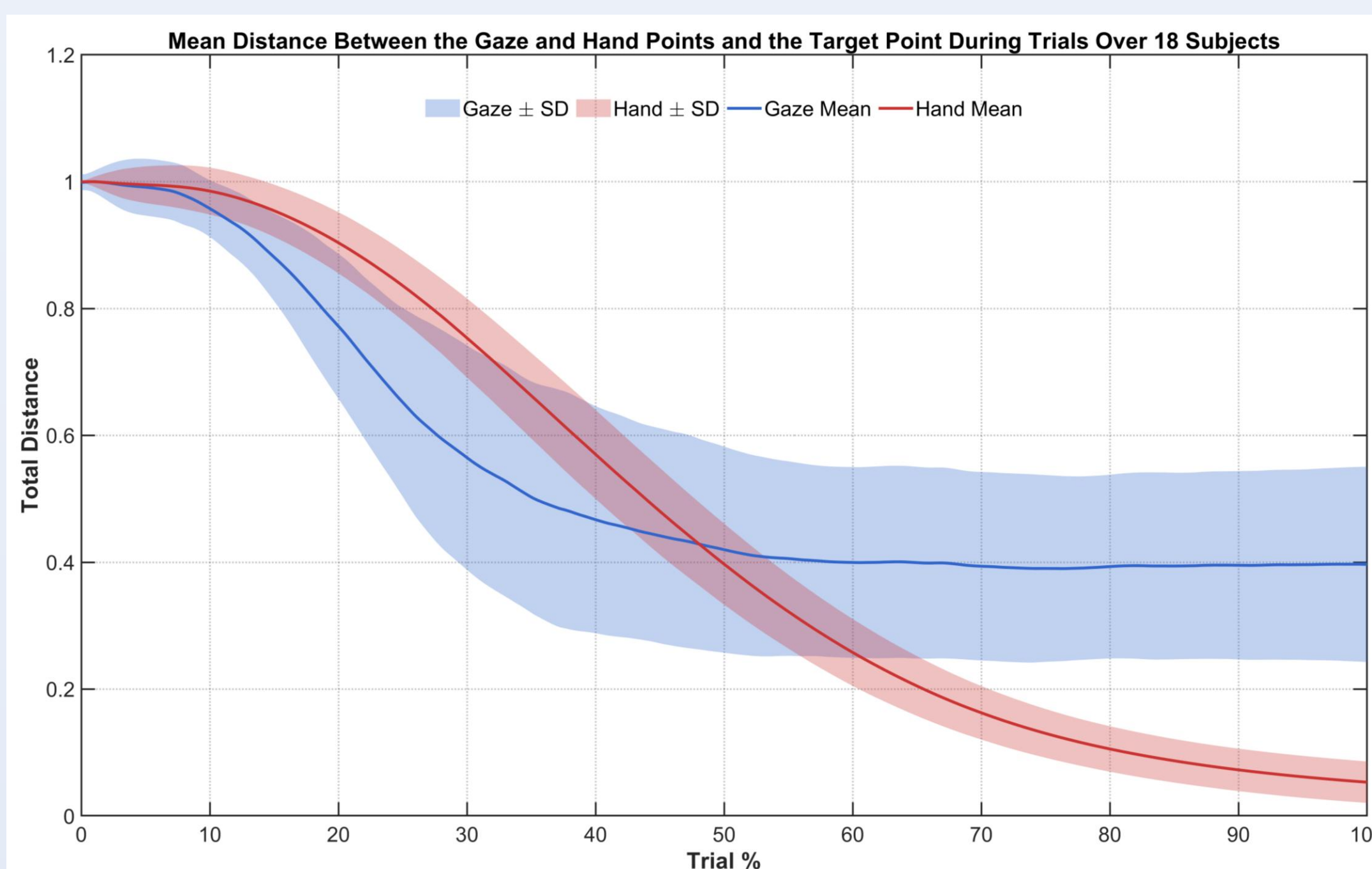


Fig 3. Difference in mean hand and gaze distances from the target

Gaze tracking may help predict reaching intent

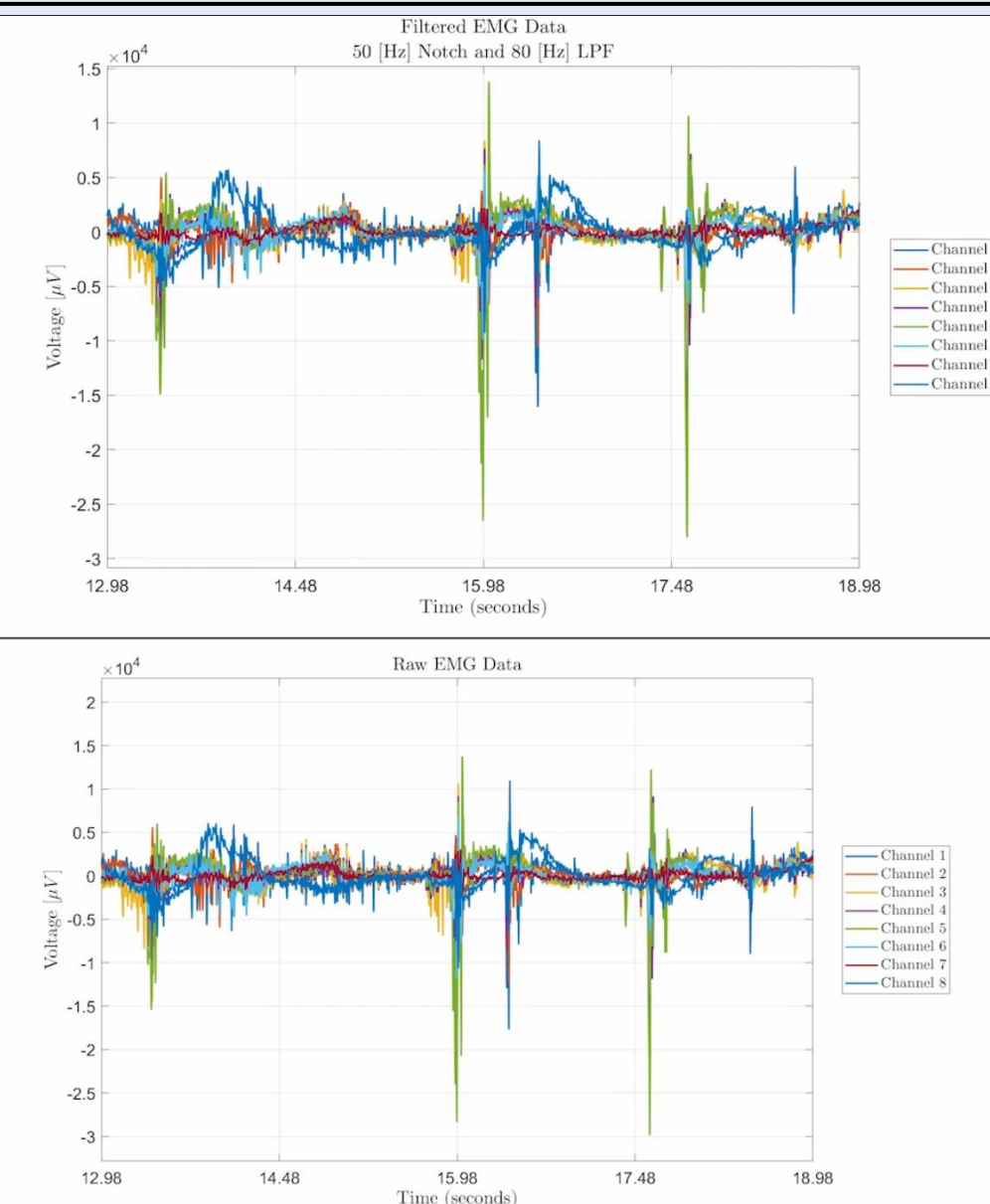
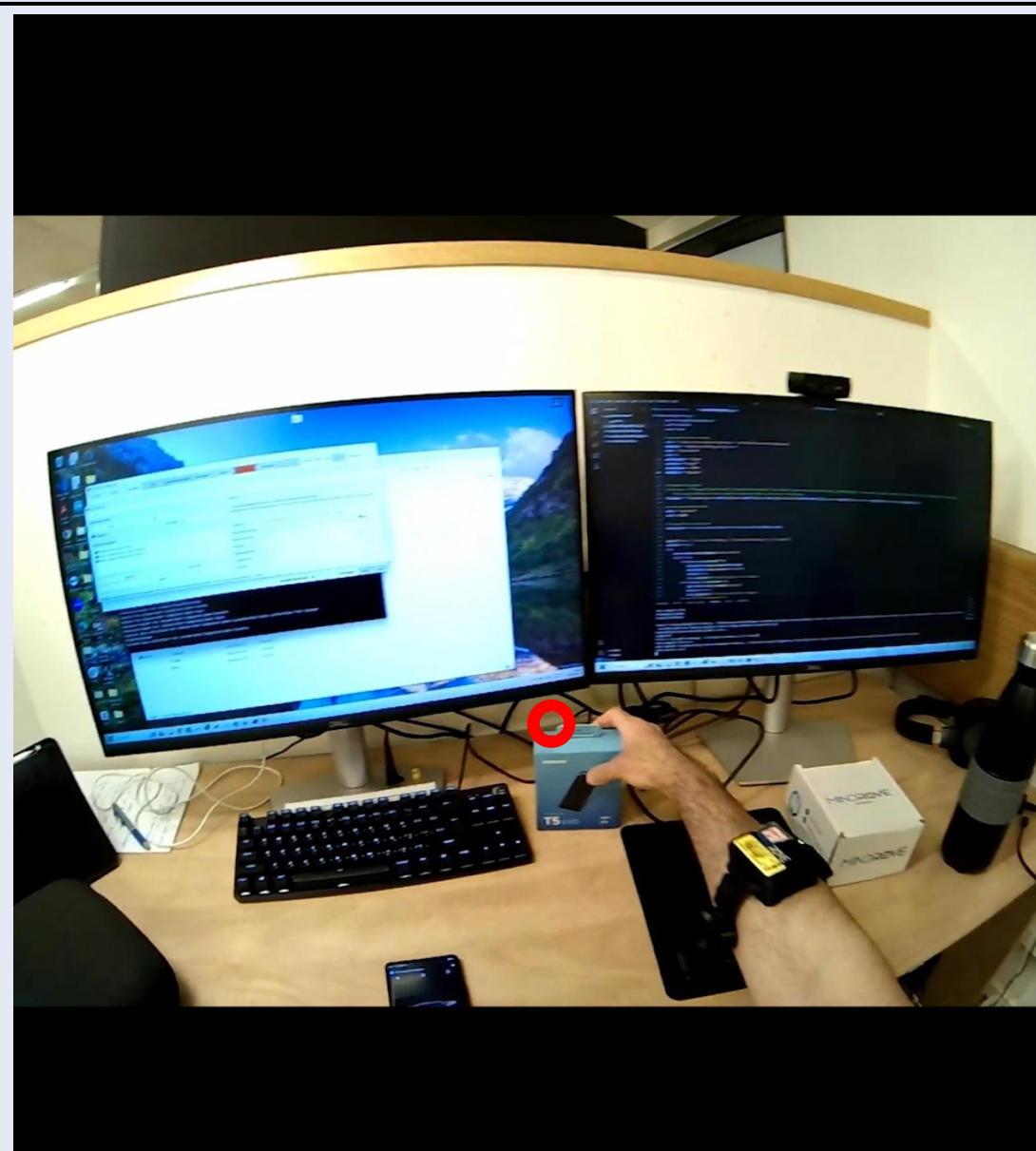


Fig 5. Screenshot from demo video showing gaze tracking and sEMG

[1] M. Kutz, *Biomedical Engineering and Design Handbook, Volume 2*, Second Edition., vol. 2. McGraw-Hill Education, 2009. Accessed: Apr. 09, 2025. [Online]. Available: <https://www.accessengineeringlibrary.com/content/book/9780071498395>
[2] N. E. Krausz, D. Lamotte, I. Batzianoulis, L. J. Hargrove, S. Micera, and A. Billard, “Intent Prediction Based on Biomechanical Coordination of EMG and Vision-Filtered Gaze for End-Point Control of an Arm Prosthesis,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 28, no. 6, pp. 1471–1480, Jun. 2020, doi: [10.1109/TNSRE.2020.2992885](https://doi.org/10.1109/TNSRE.2020.2992885).
[3] B. Lento et al., “3D-ARM-Gaze: a public dataset of 3D Arm Reaching Movements with Gaze information in virtual reality,” *Sci Data*, vol. 11, no. 1, p. 951, Aug. 2024, doi: [10.1038/s41597-024-03765-4](https://doi.org/10.1038/s41597-024-03765-4).